Ojas Mediratta

+1(470)-909-4319 | omediratta3@gatech.edu | ojasmediratta.com | linkedin.com/in/ojas-mediratta | U.S. Citizen

Education

Georgia Institute of Technology — Atlanta, GA
M.S. Robotics | *Specialization in Artificial Intelligence, Perception, and Mechanics* — *Expected May 2027*

Georgia Institute of Technology — Atlanta, GA
B.S. Computer Engineering | *Graduated with High Honors* — *May 2025*

Experience

Robotics Research Intern — May 2026 – Aug 2026 (Incoming)
GE Vernova Advanced Research — *Niskayuna, NY*
- Incoming research intern focused on autonomous industrial inspection and extreme-environment robotic systems

Graduate Research Assistant — Jan 2026 – Present
Georgia Institute of Technology - Lab for Intelligent Decision and Autonomous Robots (LIDAR) — *Atlanta, GA*
- Led tactile sensing hardware, mechanical design, and embedded-systems integration for humanoid robotics research on the Unitree G1 platform, enabling contact-aware whole-body manipulation experiments
- Supported research on diffusion-based loco-manipulation controller design for humanoid robots through data collection, hardware integration, experiment workflows, and paper authorship

Graduate Research Assistant — Aug 2024 – Present
Georgia Institute of Technology - Contextual Computing Group — *Atlanta, GA*
- Built and field-tested a custom marine robot for dolphin interaction research, integrating mechanical, electrical, firmware, and acoustic sensing subsystems across pool and ocean trials
- Designed bone-conduction headphones for researcher audio playback during pool testing and field experiments
- Developed Python audio tools for dolphin vocalization analysis, playback, and experiment preparation

Graduate Teaching Assistant — May 2025 – Present
Georgia Institute of Technology - College of Computing — *Atlanta, GA*
- Served as a teaching assistant for *Mobile and Ubiquitous Computing* and *Prototyping Intelligent Devices*; graduate-level, project based courses on embedded systems, firmware development, and edge machine learning
- Mentored 20+ students and teams in developing application prototypes and projects in robotics, IoT, and human-computer interfaces, helping refine system design, implementation, and final deliverables

Projects

Cetacean Research ROV | *C++, ESP32, Raspberry Pi, Python, Fusion, KiCAD* — Aug 2024 – Present
- Built a remotely operated vehicle (ROV) for dolphin research and enrichment, working across firmware, electronics, and mechanical design; successfully deployed in 15+ pool trials and 4 open-water trials in the Atlantic
- Designed and implemented an ESP32 firmware stack, orchestrating a cascaded PID controller, ESC-driven thrusters, internal sensors, over-the-air telemetry, and LED signaling, unifying system operation in the field
- Engineered PCBs unifying microcontroller, power, and sensor interfaces, cutting wiring volume and failure points
- Designed parts in Fusion, iterating and fabricating rapidly for waterproofing and durability for field deployment

Tactile Sensing for Humanoid Robot Manipulation | *C++, Python, Fusion, KiCAD* — Jan 2026 – Present
- Designed and fabricated flexible row-column tactile sensor arrays with Velostat pressure layers, custom electrode layouts, and robot-specific geometries for chest, forearm, gripper, and palm contact regions
- Built transfer-ready sensor mounts, wiring harnesses, and data collection setups for human demonstrations and humanoid robot testing, supporting whole-body manipulation and collaborative transport experiments

MineTrack Visual Odometry | *Python, PyTorch, Computer Vision* — Jan 2026 – Apr 2026
- Built a learning-based visual odometry pipeline with self-supervised and supervised models to estimate player motion from egocentric vision data using ground-truth pose data with Minecraft gameplay
- Trained autoencoder embeddings and multiple regression architectures for frame-to-frame SE(3) predictions
- Enhanced trajectory prediction with long-term rollout losses, reducing drift and avoiding near-zero motion collapse

Skills

Software: C, C++, Java, MATLAB, Python, Pandas, PyTorch, ROS2, Android, Kotlin
Hardware: Arduino, Raspberry Pi, ESP32, ARM, RISC-V, ESCs, Motor Controllers, Sensors
Protocols: TCP/IP, I2C, CAN, UART, SPI, Serial, USB, PWM
Developer Tools: VS Code, Arduino IDE, Android Studio, Fusion, Gazebo, KiCad, Git, Docker
Lab Tools: Oscilloscope, Multimeter, Soldering, 3D Printing, CNC Mill, Laser Cutter, Logic Analyzer
Advanced Topics: Firmware Programming, Electronics, PCB Design, Mechanical Design, Sensor Fusion